



Q-EVAC SOLUTIONS FOR
DISASTER MANAGEMENT

Quantum Optimization for Evacuation

Presented by Team NEXUS





The Evacuation Crisis

Natural disasters affect over **300 million people** annually, highlighting the urgent need for efficient evacuation solutions. Current evacuation planning methods are often slow and ineffective, resulting in increased risk due to traffic congestion and poor coordination. Real-time optimization is essential to save lives and minimize chaos during disasters, underscoring the critical role of innovative technologies like quantum computing.

Computational Complexity

Classical systems struggle with evacuation optimization due to the **exponential complexity** of combinatorial problems. With millions of variables to consider, traditional algorithms like Dijkstra's become inefficient in real-time scenarios. As the number of potential routes and conditions increases, the computational burden escalates significantly, leading to slower response times and inadequate solutions when disasters require immediate action and precise planning.



Q-Evac System



QUBO Model

Q-Evac models evacuation as a **Quadratic Unconstrained Binary Optimization (QUBO)** problem for efficient processing.

Input Data

The system utilizes crucial inputs like **road networks, population density, shelters,** and traffic data for optimization.

Output Solutions

It generates outputs for **optimal routes**, shelter allocation, and resource distribution to enhance evacuation efficiency.

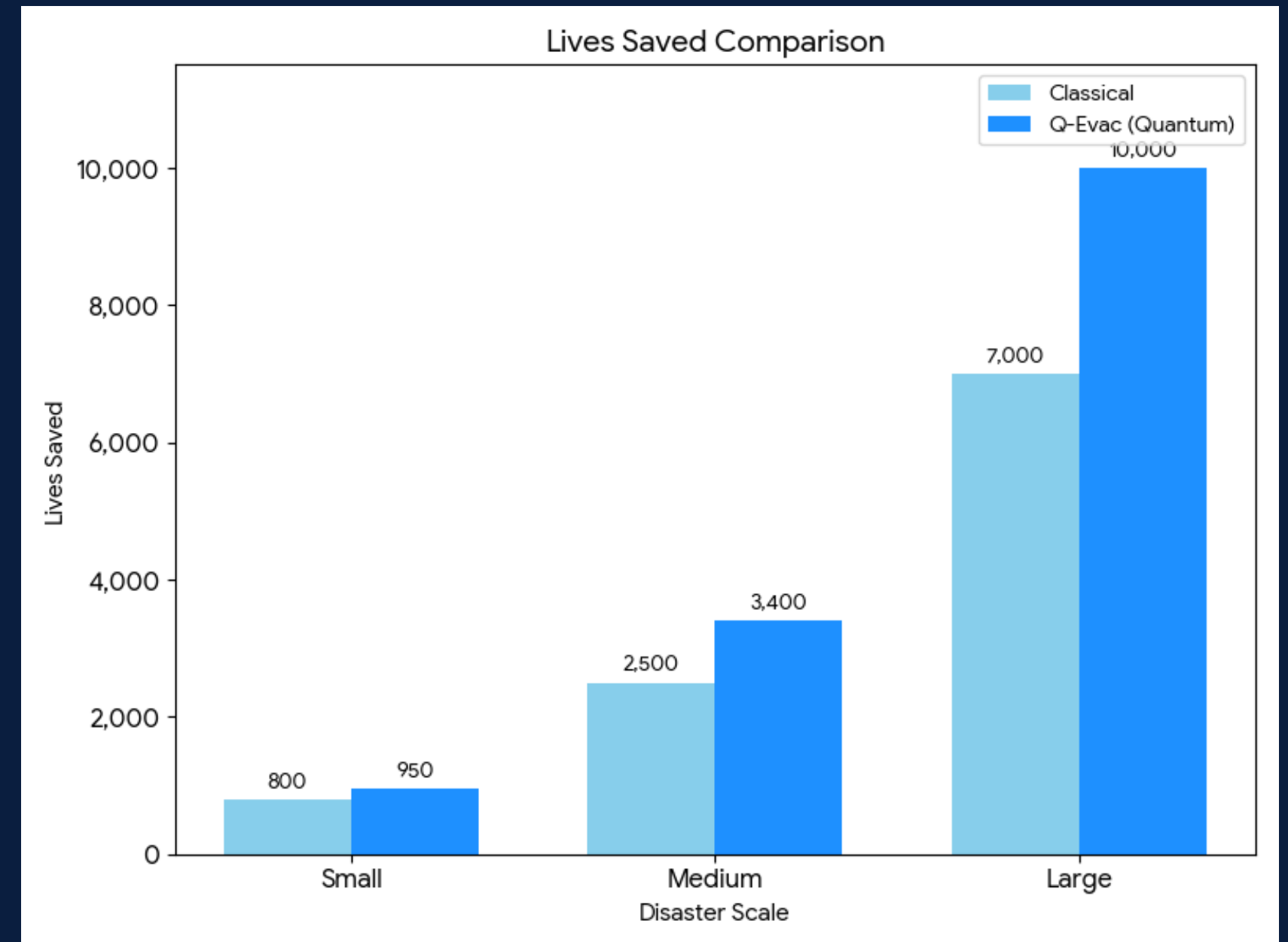
Social Impact

Lives Saved

- Faster evacuation routing reduces exposure to disaster zones.
- 20–35% reduction in evacuation time.
- Thousands of additional lives saved during large-scale disasters.
- Example Projection : Large disaster (1M affected):
- Classical systems: ~7,000 lives saved
- Q-Evac: ~10,000 lives saved → 3,000 additional lives saved

Environmental Metrics

- Reduced Traffic Congestion: Estimated Impact: 15–25% reduction in traffic congestion during evacuation.
- Lower Carbon Emissions: Projected Impact: 10–20% reduction in evacuation-related CO₂ emissions.
- Reduced Secondary Environmental Damage



Roadmap to Reality

Research & Simulation (2026–2027)



Initiating foundational research and simulations for quantum evacuation models and systems.

Pilot Testing (2028–2030)



Conducting pilot tests utilizing hybrid quantum systems for real-world disaster scenarios.

Global Deployment (2030–2035)



Full-scale deployment of Q-Evac systems across various countries and disaster agencies.

Pioneering Solutions (2036-2037)



Establishing Q-Evac as a leader in quantum disaster response technology globally.

